

AYSIN TUMAY

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Research interests: Safe offline reinforcement learning for medical device automation, with a focus on uncertainty mitigation, handling high-dimensional time series, and digital twins.

Education

University of California, San Diego

Sep. 2024 – Present

Ph.D. in Computer Science and Engineering GPA: 4.00/4.00;

La Jolla, USA

- Advisor: **Prof. Rose Yu**
- Selected coursework: Convex Optimization (CSE203B), Machine Learning Theory (CSE251C), Deep Learning (CSE251B).

Bilkent University

Sep. 2019 – Jun. 2024

B.Sc. in Electrical and Electronics Engineering GPA: 3.80/4.00, Rank: 10/158

Ankara, Turkey

- Selected coursework: Statistical Foundations of Natural Language Processing (EE486), Linear Algebra in Data Analysis and Machine Learning (EE361).

Experience

DataBoss Security and Analytics

Dec. 2022 - May 2024

Machine Learning Researcher

Ankara, Turkey

- * Developed state-of-the-art sequential models for high-dimensional time series using Gradient Boosting.

DataBoss Security and Analytics

Aug. - Sep. 2022

Machine Learning Intern

Ankara, Turkey

- * Applied Machine Learning techniques on time series data using Gradient Boosting and Neural Network models with Python.

ASELSAN, Radar and Warfare Systems

Jun. - Jul. 2022

Algorithm Design Intern

Ankara, Turkey

- * Implemented radar geolocation algorithms in MATLAB, improving detection accuracy.

UMRAM, Bilkent University

Aug. 2021

Undergraduate Research Assistant

Ankara, Turkey

- * Practiced Deep Learning methods for detecting brain illnesses in MRI scans.

Projects

Trajectory Prediction for Autonomous Driving

Apr. - Jun. 2025

- * Leveraged Transformers with a mixture of experts for trajectory prediction in the spatio-temporal domain.

Image Colorization with Convex Optimization

Jan. - Mar. 2025

- * Designed a Convolutional Neural Network model and formulated the primal and dual solutions for art image colorization.

Recipe Recommendation with Machine Learning

Sep. - Dec. 2024

- * Utilized XGBoost, SVD, BERT and similarity-based models.

Spatiotemporal Traffic Accident Prediction in Turkey

Jun. 2023- May 2024

- * Designed machine learning models to predict the probability of traffic accidents with Neural Networks and Boosting.
- * Conducted research on tackling data sparsity and spatial heterogeneity.

Wind Energy Production Prediction

Feb. - May 2023

- * Predicted hourly total electrical energy consumption in Spain with Linear Regression, Decision Tree, and AdaBoost.

A Basic Level Category Analysis with Commonsense Question Answering

Feb. - May 2023

- * Measured the common sense question answering performance of one of GPT-3.5-turbo on Family Feud language game.

Song Recommendation System for Spotify Playlists | *Python, TensorFlow*

Sep. - Dec. 2022

- * Applied unsupervised clustering algorithms such as k-Means, DBScan, and Autoencoder to give song recommendations to a playlist.

Honors & Awards

University of California, San Diego, CSE Graduate Student Researcher Fellowship * Full tuition waiver, and stipend for entire duration of Ph.D. studies.	Mar. 2024
University of California, San Diego, ECE Fellowship * Recipient of first year's financial support rewarded to exceptional PhD applicants.	Mar. 2024
University of Illinois at Urbana-Champaign, the Dilip and Sandhya Sarwate Graduate Fellowship * Awarded to outstanding incoming graduate students in the area of communications.	Mar. 2024
University of Illinois at Urbana-Champaign, the Promise of Excellence Fellowship * Given to incoming PhD students who demonstrate the potential to achieve great things in the ECE field.	Mar. 2024
3rd Place at Ipsos Datathon Jupyter * Solved a case study about predicting a company's market share by trend analysis using ARIMA and Linear Regression.	May 2023
5th Place at Invent Analytics Data Analysis Challenge Jupyter * Trained and tested a Machine Learning model to forecast the sales amount of a clothing brand.	Sep. 2022

Technical Skills

Languages: Python, VHDL, MATLAB, Assembly 8051,
Tools: VSCode, Pycharm, Jupyter Notebook, Linux, GitHub, LaTeX, MS applications,
AI/ML Frameworks: Gymnasium, Pytorch, Tensorflow, Scikit-learn,
Electronics: LTSpice, DipTrace, Proteus, MCU IDE.

Publications

A. Tumay, S. H. Sun, S. Fereidooni, A. Dumas, E. Jortberg, R. Yu. “Guardian-regularized Safe Offline Reinforcement Learning for Smart Weaning of Mechanical Circulatory Devices.” ***Machine Learning for Health 2025*** (Under review).
S. H. Sun, **A. Tumay**, S. Fereidooni, A. Dumas, E. Jortberg, R. Yu. “Probabilistic Digital Twin for Data-driven Smart Weaning of Medical Circulatory Devices.” ***NeurIPS Learning from Time Series for Health 2025*** (Under review).
A. Tumay, M. E. Aydin, A. T. Koc, S. S. Kozat. “Hierarchical Ensemble-based Feature Selection for Time Series Forecasting.” 2023. DOI: 10.48550/ARXIV.2310.17544.

Professional Service

- Reviewer for ML4H symposium (2025).
- Reviewer for NeurIPS TS4H workshop (2025).

Extracurricular

- Organizing lunch events for Women in Data Science at UCSD.
- Active member at Young Entrepreneurs Society, and IEEE Student Branch, 2019-2020.
- Ankara Start-up Summit committee member, 2019-2020.
- Classical guitar player in high school orchestra, 2018.